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Exercise 2 : SQL Aggregate Functions & Operators

1) `SELECT DISTINCT (department)
FROM students ;`

Output table

department
IT
HR
Finance

2) `SELECT department,
AVG(age) AS avg-age
FROM students
GROUP BY department ;`

Output table :

department	avg-age
IT	20,5
HR	22
Finance	23

3)
 SELECT department ,
 COUNT (student_id) AS student_count
 FROM students
 GROUP BY department
 HAVING student_count > 1 ;

Output Table

department	student_count
IT	2
HR	2

4)
 SELECT student_id ,
 name ,
 age ,
 department
 FROM students
 WHERE age BETWEEN 21 AND 23 ;

Output Table

student_id	name	age	department
2	^{Bob} Alice	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5) SELECT student_id,
 name,
 age,
 department
 FROM students
 WHERE (department = 'IT' OR department = 'HR')
 AND age > 21 ;

Output Table

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6) ~~SELECT department,
 SUM(total) AS~~
 SELECT department
 SUM(credits) AS total_credits
 FROM courses
 GROUP BY department
 HAVING total_credits > 5 ;

Output Table :

department	total_credits
IT	11

~~Finance~~
~~HR~~

```

7) SELECT      course_id,
               course_name,
               department,
               credits
FROM    courses
WHERE   NOT credits = 4 ;

```

Output Table

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

```

8) SELECT      course_id,
               course_name,
               credits
FROM    courses
ORDER BY credits DESC
LIMIT 3 ;

```

Output Table :

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basic	3

9) ~~SELECT MAX(grade) AS max_grade,~~
~~MIN(grade) AS min_grade,~~
~~Avg(grade) AS avg_grade~~
 SELECT MAX(grade) AS max_grade,
 MIN(grade) AS min_grade,
 AVG(grade) AS avg_grade
 FROM enrollments;

Output Table

max_grade	min_grade	avg_grade
90	78	84.16

10) SELECT course_id,
~~count~~ COUNT(enrollment_id) AS enrollment_count
 FROM enrollments;

Output Table

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

11) SELECT department,
SUM(salary) AS total_salary,
SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

12) SELECT department,
AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING avg_salary > 55 000 ;

Output Table :

department	avg_salary
IT	61 000
Finance	70 000

13) SELECT employee_id
 name
 salary
 bonus
 SUM(salary + bonus) AS total_compensation
 FROM salaries
 WHERE total_compensation > 60000 ;

Output Table

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

14) SELECT department
SUM(budget) AS total-budget,
AVG(budget) AS avg-budget
FROM projects
GROUP BY department
HAVING avg-budget > 70000;

Output Table

department	total-budget	avg-budget
IT	270 000	135 000
Finance	80 000	80 000

15) SELECT project-id,
project-name,
department,
budget
FROM projects
WHERE NOT department = 'Marketing'
~~AND budget BETWEEN 50000 AND 120000;~~
AND budget BETWEEN 50000 AND 120000;

Output Table

project-id	project-name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
3	Dashboard	IT	150 000
5	HR Portal	HR	50 000